Felix Jimenez

Website: felix-jimenez.com
Email: felix-jimenez@tamu.edu
LinkedIn: felix-jimenez-557478b1
GitHub: github.com/feji3769
Work Authorization: US Citizen

Education

Texas A&M University — College Station, TX
Ph.D. in Statistics — 2020–Present

University of Colorado Boulder — Boulder, CO
M.S. in Applied Mathematics — 2017–2018

University of Colorado Boulder — Boulder, CO
B.S. in Applied Mathematics — 2014–2018

Experience

Texas A&M University — College Station, TX
Research Assistant — 2020-Present

- Vecchia Approximation for Bayesian Optimization
- Using Gaussian process approximation for Bayesian optimization.

National Institute of Standards and Technology (NIST) — Boulder, CO
Mathematical Statistician — 2019-2020

- Hierarchical Gaussian Processes for Outlier Detection
- Created an outlier detection procedure using Gaussian processes.
- Permutation Test for Forensic Analysis of Glass
- Developed a permutation test for μ-XRF spectra to determine if two sets of glass come from the same source.
- Generative Adversarial Networks on Low Dimensional Data
- Investigated the impact of data quality and complexity on fidelity of samples generated by GANs.

National Institute of Standards and Technology (NIST) — Boulder, CO
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University of Colorado Boulder/NIST — Boulder, CO
Professional Research Assistant — 2018-2019

- See Above.

University of Colorado Boulder — Boulder, CO
Research Assistant — 2015-2018

- Statistics Consulting Lab
- Acted as consultant for groups in Environmental Engineering, Mechanical Engineering and Academic Services.
- Applied Math Department
- Undergraduate research assistant for project between physics and applied math.

Publications

[1] **F. Jimenez**, A. Koepke, and M. Frey, "Generative Adversarial Network Performance for Low Dimensional Data", Submitted to NIST Journal of Research.

[2] A. Koepke, **F. Jimenez**, K. Kroenlein, and C. Muzny, "Hierarchical Bayesian change point models for chemical properties inference", presented at the 2019 Joint Statistical Meetings, 2019.

[3] J. Splett, A. Koepke, and **F. Jimenez**, "Estimating the Parameters of Circles and Ellipses Using Orthogonal Distance Regression and Bayesian Errors-in-Variables Regression", presented at the 2019 Joint Statistical Meetings, 2019.

[4] K. Tucker, B. Zhu, R. J. Lewis-Swan, J. Marino, **F. Jimenez**, J. G. Restrepo, and A. M. Rey, "Shattered time: Can a dissipative time crystal survive many-body correlations?", *New J.Phys.*, vol. 20, Dec. 2018.

Teaching

- **Teaching Assistant** at Texas A&M University — Fall 2020
  Statistical Learning (Stat 436)
- **Teaching Assistant** at University of Colorado Boulder — Fall 2017
  Calculus 1 (APPM 1350)
- **Short Course Instructor** at University of Colorado Boulder — Fall 2017 - Spring 2018
  Intro. to R and Social Network Analysis
- **Learning Assistant** at University of Colorado Boulder — 2016-2017
  Matrix Methods and Physics I
- **Instructing Assistant with SASC** at University of Colorado Boulder — 2015-2016
  College Algebra and Intro. to College Math

Programming Languages

- **Advanced:** Python, R.
- **Proficient:** C++.
- **Beginner:** Java, Scala.

Tools & Libraries

- **Advanced:** Numpy, ggplot, Pytorch.
- **Intermediate:** Linux, Spark, Slurm, scikit-learn.
- **Beginner:** Bash, SQL, AWS.

Talks

- Hierarchical GPs for Outlier Detection — 2020
  JSM, Topic Contributed
- Glass Stats and Crime: Let's Make the Right Decision — 2019
  JSM, Topic Contributed
- Generative Adversarial Network Performance for Low Dimensional Data — 2019
  QRPC, Topic Contributed
- Generative Adversarial Network Performance for Low Dimensional Data — 2019
  NIST Colloquium

Posters

- Measuring Inter-operator Variability — 2016
  Rocky Mountain Fluid Mechanics Symposium
- Interacting Quantum Dipoles — 2016
  Dynamics Days

Scholarships and Awards

- Lechner Graduate Grant, \$6,000 2020
- Cash-In-Your-Account Award, \$200 2020
- Biological Sciences Initiative, \$700 2016
- Biological Sciences Initiative, \$3,500 2015